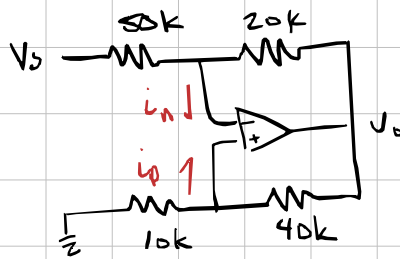


① Back 54

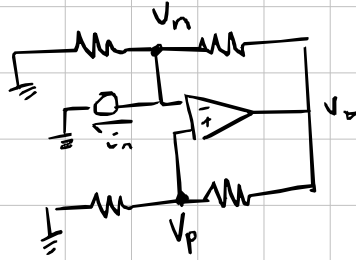


$$I_B = -100 \text{ nA}$$

$$I_{os} = \pm 10 \text{ nA}$$

Superposition:

i_n :



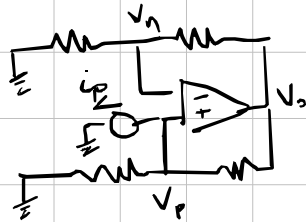
$$V_n = V_p; \quad V_p = \frac{10}{40+10} V_o = \frac{V_o}{5}$$

$$i_n = \frac{0 - V_n}{50k} + \frac{V_o - V_n}{20k} = \frac{V_o}{20k} - \frac{7V_n}{100k}$$

$$i_n = \frac{V_o}{20k} - \frac{7V_o}{500k} = 3.6 \times 10^{-5} V_o$$

$$V_o = 27.7k i_n$$

i_p :



$$V_n = \frac{50k}{20k+50k} V_o = \frac{5}{7} V_o$$

$$i_p = \frac{0 - V_p}{10k} + \frac{V_o - V_p}{40k} = \frac{V_o}{40k} - \frac{5V_p}{40k}$$

$$i_p = \frac{V_o}{40k} - \frac{25V_o}{280k} = -6.4 \times 10^{-5} V_o$$

$$V_o = -15.5k i_p$$

$$V_o = 27.7k i_n - 15.5k i_p$$

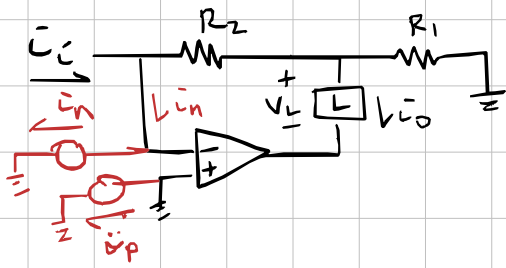
$$= 27.7k \left(I_B + \frac{I_{os}}{2} \right) - 15.5k \left(I_B - \frac{I_{os}}{2} \right)$$

$$= 12.2k I_B + 2.16k I_{os} = -1.2 \text{ mV} \pm 0.216 \text{ mV}$$

Error bound: $V_{OE \text{ min}}: -1.005 \text{ mV}$

$V_{OE \text{ max}}: -1.438 \text{ mV}$

② Book 5.10 - 6050 only



i_p has no effect as there is a direct path to ground.

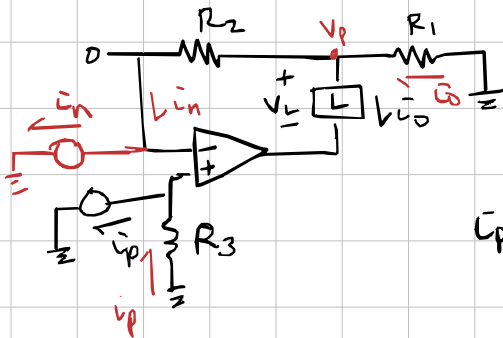
i_n has same gain as i_c

(error due to i_n)

$$\bar{i}_o = -\left[1 + \frac{R_2}{R_1}\right] \bar{i}_n \quad (\text{from eq. 2.17})$$

$$\bar{i}_o = -\left[1 + \frac{R_2}{R_1}\right] (I_B + I_{os}/2) \quad \text{No fix}$$

fix: Add dummy resistor to + term:



Via superposition!
already know i_n contrib.

i_p causes $V_p = -i_p R_3$

$V_n = V_p$, no current through R_2 :

$$0 - V_p = \bar{i}_o R_1 = i_p R_3$$

$$\bar{i}_o = \frac{R_3}{R_1} i_p$$

$$\bar{i}_o = \frac{R_3}{R_1} i_p - \left[1 + \frac{R_2}{R_1}\right] \bar{i}_n$$

$$= \frac{R_3}{R_1} (I_B - I_{os}/2) - \left[1 + \frac{R_2}{R_1}\right] (I_B + I_{os}/2)$$

$$= \left(\frac{R_3}{R_1} - \left[1 + \frac{R_2}{R_1}\right] \right) I_B - \left(\frac{R_3}{R_1} + \left[1 + \frac{R_2}{R_1}\right] \right) \frac{I_{os}}{2}$$

can minimize due to - sign!

$$\frac{R_3}{R_1} - 1 - \frac{R_2}{R_1} = 0 \rightarrow R_3 = R_1 + R_2$$

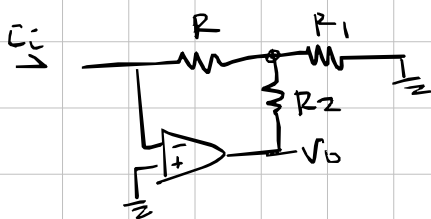
$$\bar{i}_o = 0 I_B - \left(\frac{R_1 + R_2}{R_1} + \frac{R_1 + R_2}{R_1} \right) \frac{I_{os}}{2}$$

fix with $R_3 = R_1 + R_2$:

$$\bar{i}_o = -\left(1 + \frac{R_2}{R_1}\right) I_{os}$$

sign does not matter

③ Book 5.20



$$V_o = -k R i_c \quad k = 1 + \frac{R_2}{R_1} + \frac{R_2}{R}$$

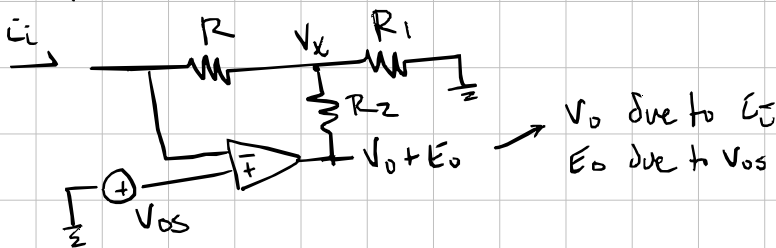
$$R = 500k, R_1 = 1k, R_2 = 99k$$

$$A = 100dB, PSRR = 80dB \quad V_{OS0} = 0$$

$$\pm V_s = \pm (5 + 1 \sin(2\pi t)) V$$

$$k = 100.2 \rightarrow V_o = -50M i_c$$

Input V offset model:



V_o due to i_c
 E_o due to V_{OS}

V_o contribution from V_{OS} : $V_p = V_{OS} \quad V_n = V_{OS}$

No current through R ($i_c = 0$ - superposition)

$$\therefore V_x = V_{OS}$$

$$V_{OS} = E_o \frac{R_1}{R_1 + R_2} \rightarrow E_o = \left(1 + \frac{R_2}{R_1}\right) V_{OS}$$

V_{OS} contributions:

$$a: V_{OS, max} \approx 5V, A = 100dB = 100k$$

$$\rightarrow V_{OS, a} = \frac{5}{100k} = 50\mu V$$

PSRR: ΔV_s is sine wave portion; $\Delta V_s = 1 \sin 2\pi t$

$$PSRR = 80dB = 10k$$

$$\rightarrow \Delta V_s = \pm 1$$

$$\rightarrow V_{OS, PSRR} = \frac{1}{10k} = 100\mu V$$

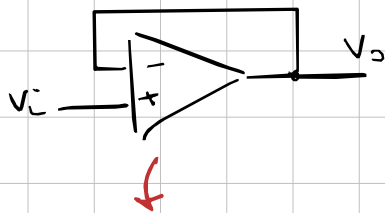
$$a) V_{OS} = 50\mu + 100\mu = 150\mu V$$

$$E_o = \left(1 + \frac{R_2}{R_1}\right) V_{OS} = 15mV$$

$$b) V_o = A_{ideal} i_c + E_o = -50 \frac{V}{\mu A} i_c \pm 15mV$$

Book solution leaves sin in this such that
 $E_o = \pm (5mV + 10mV \sin 2\pi t)$

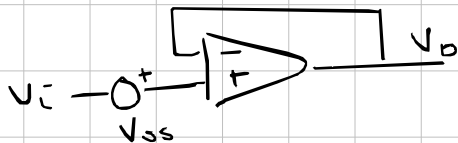
④ Book 5.27



$$a_{min} = 10^4 \text{ V/V} \quad V_{os0(max)} = 2 \text{ mV}$$

$$\underline{CMRR_{0dB(min)} = PSRR_{0dB(min)} = 74 \text{ dB}}$$

min/max are all worst case values



view virtual short! $V_{cm} = V_i$

$\therefore V_{o(max)} \approx V_i$ due to voltage follower

$\therefore E_o = V_{os}$ (voltage follower)

a) $V_i = 0 \text{ V}$, no power supply changes

$$\therefore V_{o(max)} = 0, \quad V_{cm} = 0$$

only error contribution from V_{os0}

$$\rightarrow V_{os} = V_{os0} = 2 \text{ mV}$$

$$\rightarrow \boxed{E_o = 2 \text{ mV}}$$

b) $V_i = 10 \text{ V}$, no power supply changes

$$\therefore V_{o(max)} = 10 \text{ V}, \quad V_{cm} = 10 \text{ V}$$

$$V_{os} = V_{os0} + \frac{V_{cm}}{CMRR} + \frac{V_{o(max)}}{a}$$

$$CMRR = 74 \text{ dB} = 5012$$

$$V_{os} = 2 \text{ mV} + 2 \text{ mV} + 1 \text{ mV} = 5 \text{ mV}$$

$$\boxed{E_o = 5 \text{ mV}}$$

c) All effects from above AND $\Delta V_s = 15 - 12 = 3 \text{ V}$

$$PSRR = 74 \text{ dB} = 5012$$

$$V_{os} = 5 \text{ mV} + \frac{\Delta V_s}{PSRR}$$

$$= 5 \text{ mV} + 0.6 \text{ mV} = 5.6 \text{ mV}$$

$$\boxed{E_o = 5.6 \text{ mV}}$$